4.1 Rate of change and secants

Key point · Definition and results. For $a \neq b$ in the domain of f, the rate of change between a and b is $\dfrac{f(b)-f(a)}{b-a}$. It is the slope of the secant through the corresponding points. In context, the unit of the rate is the unit of f divided by that of the variable.

Status and justification. Definition; formula for the slope of a line, already studied. With $b = a+h$, $h \neq 0$, the rate becomes $\dfrac{[f(a+h)-f(a)]}{h}$.

Example 1. For $f(x) = x^2$, between 1 and 3, the rate is $\dfrac{9-1}{3-1} = 4$. Between 1 and $1+h$, it is $2+h$. The slopes for $h = 0.1$ and $h = -0.1$ are 2.1 and 1.9.

Watch out. The rate is not $\dfrac{f(b)}{f(a)}$; h represents a gap, positive or negative depending on the domain.

Application. Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2+h$.

4.2 Derivative at a point

Key point · Definition and results. f is differentiable at a when the rate $\dfrac{[f(a+h)-f(a)]}{h}$ tends to a real number as h tends to 0, within the domain. This real number is $f\prime(a)$. An intuitive approach to the limit is used here. In kinematics, if f gives a position in metres and time is in seconds, the derivative at a point represents an instantaneous speed in $\mathrm{m\,s^{-1}}$.

Status and justification. For $f(x) = x^2$, the rate at a is $2a+h$, which gets closer to $2a$; hence $f\prime(a) = 2a$. This calculation will be taken up again as a derivative function in chapter 6.

Example 2. For $s(t) = t^2 + 3t$, the rate around 2 is $7+h$, so $s\prime(2) = 7\ \mathrm{m\,s^{-1}}$. The average speed on $[2,3]$ is $8\ \mathrm{m\,s^{-1}}$; it is not the speed at time 2.

Watch out. We do not replace h by 0 in an unsimplified quotient: we would get a division by 0.

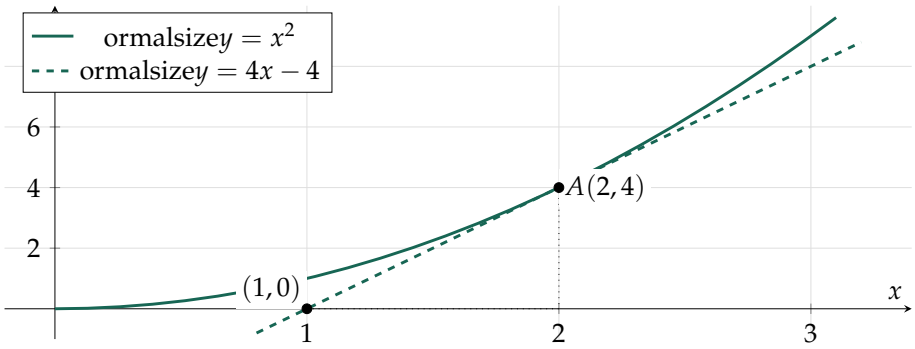
Application. For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

4.3 Tangent and affine approximation

Key point · Definition and results. If f is differentiable at a, the tangent at the point $(a, f(a))$ has equation $y = f(a) + f\prime(a)(x-a)$. For small h, we use $f(a+h) \approx f(a) + f\prime(a)h$. This local approximation does not by itself provide a bound on the error.

Status and justification. The tangent has slope $f\prime(a)$ and passes through $(a, f(a))$. In $y = f\prime(a)x + p$, substituting this point gives $p = f(a) - af\prime(a)$, hence the equation.

Example 3. For $f(x) = x^2$ at 2, tangent $y = 4 + 4(x-2) = 4x - 4$. We approximate 2.01^2 by $4 + 4(0.01) = 4.04$; the exact value is 4.0401, error $0.0001 = h^2$.



The tangent at A also passes through $(1,0)$. Its slope is $\dfrac{4-0}{2-1} = 4$. The dashed line distinguishes it from the curve, even in black and white.

Watch out. A tangent may cross the curve again elsewhere. "Only one common point" is not a definition.

Application. Given $f(3) = 5$ and $f'(3) = -2$. Write the tangent and approximate $f(3.04)$.

4.4 When differentiability fails

Key point · Definition and results. The square root function is defined on $[0, +\infty)$ but is not differentiable at 0. The absolute value function is defined on $\mathbb{R}$ but is not differentiable at 0. Being defined, or having a continuous curve, therefore does not guarantee differentiability.

Status and justification. For $h > 0$, $\dfrac{\sqrt{h} - 0}{h} = \dfrac{1}{\sqrt{h}}$, which does not tend to a finite real number as h gets closer to 0. For $|x|$, the rate $\dfrac{|h|}{h}$ equals 1 on the right and -1 on the left: no common value.

Example 4. For $|x|$, the secants from the origin all have slope 1 on the right and -1 on the left. The corner point does not have a unique tangent with finite slope. For the square root, the limiting direction is vertical; there is no real derivative.

Watch out. A very large slope is not a proof of non-differentiability; the behaviour of the rate must be explained.

Application. Justify that $g(x) = |x-2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate.